# Last-Mile Delivery Operations Analytics

MySQL Business Analytics Project

## Sprint 1: Business Understanding and Data Understanding

### 1.1 Company Background

The company operates as a logistics and delivery service provider, where orders need to be assigned, transported, and delivered to customers across different delivery zones. Understanding the business context helped establish how customers, orders, delivery zones, drivers, vehicles, and delivery outcomes are connected. The analysis also considered key operational factors such as service types, order priorities, delivery status, delivery distance, delivery duration, and multiple delivery attempts. This business understanding provided the foundation for the subsequent data analysis by helping identify the key operational and customer-related questions that the project aims to answer, such as understanding demand, customer behaviour, delivery performance, resource utilization, and delivery problems.

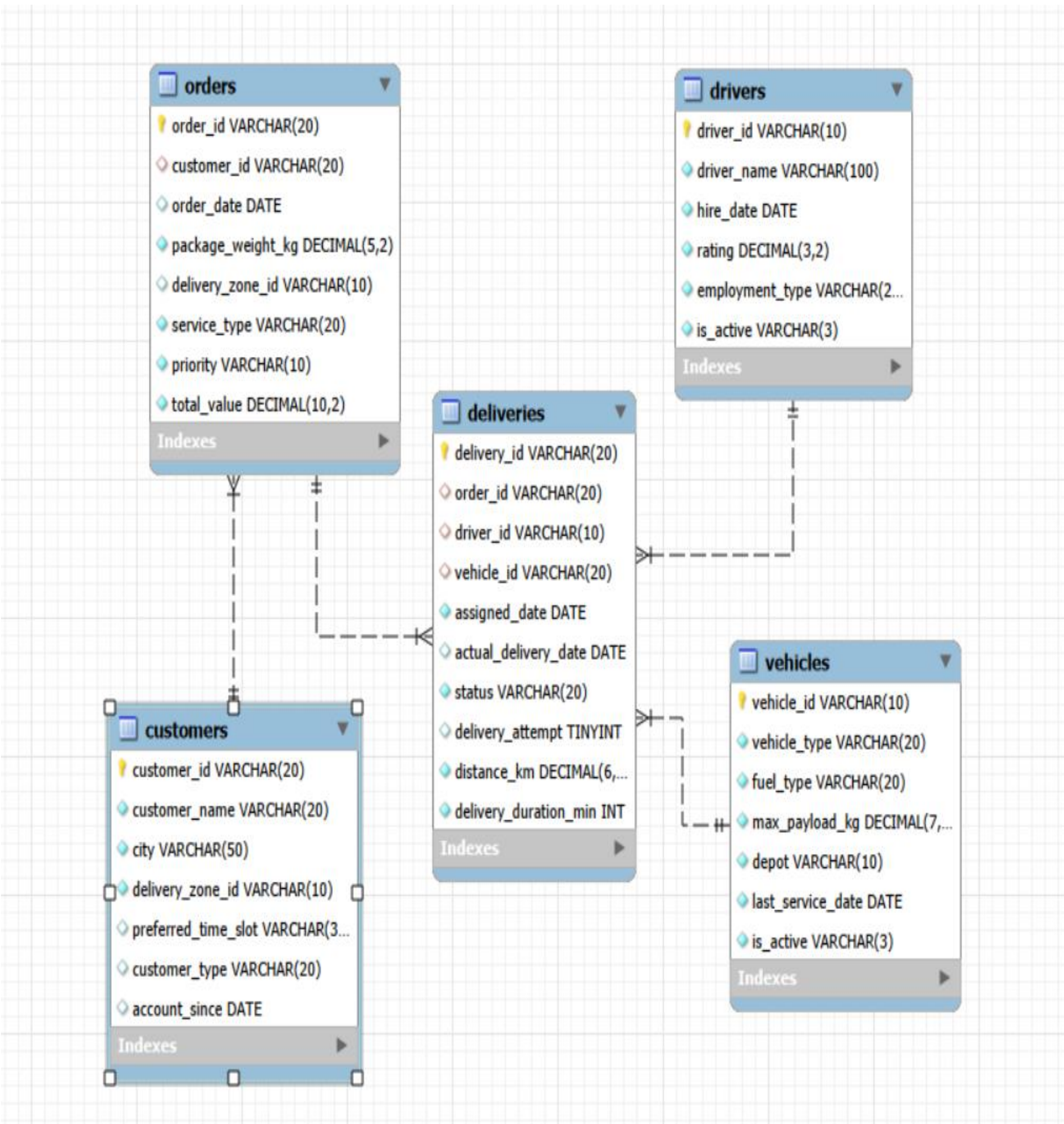
### 1.2 Role

In this project, I assumed the role of a Data Analyst supporting the Operations team. My responsibility was to analyse the available business data, understand operational patterns, and answer key business questions using SQL-based analysis. The analysis focused on areas such as delivery demand, customer order behaviour, delivery performance, driver and vehicle performance, and delivery problems. The insights generated from the analysis are intended to support the Operations team in identifying trends, understanding operational challenges, and making data-driven decisions to improve last-mile delivery operations.

### 1.3 Understanding the ER Diagram

The database is designed to represent the key entities involved in the last-mile delivery process. The main entities are Customers, Orders, Deliveries, Drivers, and Vehicles. The customers table stores customer information and is connected to the orders table through customer_id, allowing customer ordering behaviour to be analyzed. The orders table contains information such as order date, package weight, delivery zone, service type, priority, and total order value. Each order is associated with delivery information through the deliveries table using order_id.

The deliveries table acts as the central operational table and records delivery-related information such as the assigned driver, assigned vehicle, assigned date, actual delivery date, delivery status, number of attempts, distance, and delivery duration. The drivers table stores driver details such as driver name, hire date, rating, employment type, and active status, while the vehicles table contains vehicle type, fuel type, maximum payload, depot, service date, and active status. The relationships between these tables allow the analysis to connect customers and orders with delivery performance, drivers, and vehicles, providing a structured foundation for operational analysis.



## 1.4 Analytical Thinking from the ER Diagram

Before writing SQL queries, the ER diagram was used to understand how the available data is structured and how different business questions could be investigated. The required tables, columns, relationships, and comparison criteria were identified based on the business situation.

**1.** Management wants to find customers who have placed multiple orders. What information would you need to identify them?

I would need the orders table, specifically customer_id and order_id. Customers can be grouped using customer_id, and the number of orders placed by each customer can be determined by counting their orders. Customers with more than one order can then be identified as customers who have placed multiple orders.

Tables: orders

**2**. The Operations team wants to identify orders that required more than one delivery attempt. Where would you find the information needed to investigate this?

I would use the deliveries table, specifically order_id and delivery_attempt. The order_id identifies the order, while delivery_attempt indicates the number of delivery attempts. Orders where the recorded delivery attempt is greater than one can be identified as orders that required multiple delivery attempts.

Table: deliveries
Relationship: deliveries.order_id connects to orders.order_id if additional order information is required.

**3.** The Customer team wants to compare Business and Individual customers based on their ordering activity. Which tables and columns would you need?

I would need the customers and orders tables. The customer_type column from the customers table can be used to distinguish between Business and Individual customers. The customer_id column can be used to connect customers with their orders. From the orders table, I would examine metrics such as the number of orders, total order value, priority, and delivery zone to compare the ordering behaviour of the two customer types.

Tables: customers, orders
Relationship: customers.customer_id → orders.customer_id

**4.** The Operations team wants to compare different service types based on how far deliveries travel and how long they take. Which tables would you need to connect?

I would connect the orders and deliveries tables using order_id. The service_type column from the orders table would be compared with distance_km and delivery_duration_min from the deliveries table. This would allow the Operations team to examine whether different service types have different travel distances and delivery durations.

Tables: orders, deliveries
Relationship: orders.order_id → deliveries.order_id

**5.** The team wants to identify which drivers have handled deliveries and examine their recorded ratings. What information would you need?

I would use the drivers and deliveries tables. The driver_id column can be used to connect the two tables and identify which drivers have handled deliveries. The driver_name and rating columns from the drivers table provide the driver's details and recorded rating, while delivery_id or order_id from the deliveries table can be used to determine the number of deliveries handled by each driver.

Tables: drivers, deliveries
Relationship: drivers.driver_id → deliveries.driver_id

**6**. Operations wants to understand whether different types of vehicles are being used for different deliveries. Which tables and columns would you need?

I would use the vehicles and deliveries tables. The vehicle_id column connects the two tables. The vehicle_type and fuel_type columns from the vehicles table can be used to identify different types of vehicles, while the delivery_id, order_id, and delivery-related columns from the deliveries table can be used to examine the deliveries handled by each vehicle type. This would help determine how different vehicle types are being utilized across deliveries.

Tables: vehicles, deliveries
Relationship: vehicles.vehicle_id → deliveries.vehicle_id

**7.** Management wants to compare delivery performance across different delivery zones. What information would you need from the database?

I would need the orders and deliveries tables because the delivery zone is stored in the orders table, while delivery performance information is stored in the deliveries table. The delivery_zone_id from orders can be connected to deliveries through order_id. I would examine delivery status, delivery duration, delivery attempts, and distance across different delivery zones to compare their performance.

Tables: orders, deliveries

**8.** Operations wants to investigate whether heavier packages are associated with longer delivery durations. Which information would you need, and where would you find it?

I would need the orders and deliveries tables. The package_weight_kg column in the orders table provides the package weight, while delivery_duration_min in the deliveries table provides the delivery duration. The two tables can be connected using order_id. I would compare package weight with delivery duration to determine whether heavier packages are associated with longer delivery times.

Tables: orders, deliveries
Relationship: orders.order_id → deliveries.order_id

**9.** Management wants to understand whether delivery outcomes differ across service types. What tables and information would you bring together?

I would connect the orders and deliveries tables using order_id. The service_type column from the orders table identifies the type of delivery service, while the status column from the deliveries table represents the delivery outcome. I would compare outcomes such as Delivered, Failed, Pending, and Rescheduled across the different service types.

Tables: orders, deliveries
Relationship: orders.order_id → deliveries.order_id

**10.** Operations wants to understand which customers have placed orders and how their orders are being handled. What information would you need to connect a customer with their orders and deliveries?

I would need the customers, orders, and deliveries tables. The customer_id connects customers with their orders, while order_id connects orders with their deliveries. From the customers table, I would use customer information such as customer_id, customer_name, and customer_type. From the orders table, I would examine order details such as order_date, service_type, priority, delivery_zone_id, and total_value. From the deliveries table, I would examine how the orders were handled using information such as driver_id, vehicle_id, status, delivery_attempt, distance_km, and delivery_duration_min.

Tables: customers, orders, deliveries